

8.012 Problem Set 9

Nilangshu Sarkar

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Problem 1

Kleppner & Kolenkow, Problem 7.5

7.5 When an automobile rounds a curve at high speed, the loading (weight distribution) on the wheels is markedly changed. For sufficiently high speeds the loading on the inside wheels goes to zero, at which point the car starts to roll over. This tendency can be avoided by mounting a large spinning flywheel on the car.

a. In what direction should the flywheel be mounted, and what should be the sense of rotation, to help equalize the loading? (Be sure that your method works for the car turning in either direction.)

b. Show that for a disk-shaped flywheel of mass m and radius R , the requirement for equal loading is that the angular velocity of the flywheel, ω , is related to the velocity of the car v by

$$\omega = 2v \frac{ML}{mR^2},$$

where M is the total mass of the car and flywheel, and L is the height of the center of mass of the car (including the flywheel) above the road. Assume that the road is unbanked.

a.

Let the Car move in the xy Plane. So the Torque of turning is along the z Axis.

During a turn, lateral friction forces acting below the center of mass create a roll torque that tends to tip the car. The ground constraint prevents this rotation, causing a redistribution of normal forces between the inner and outer wheels. The resulting torque from the unequal normal forces balances the roll

torque produced by friction. The net normal forces are always equal to the weight Mg . So the last limit of normal-force balancing is zero on inner wheels and Mg on outer wheels. Beyond this point, the torque from lateral friction can no longer be balanced by normal-force redistribution, so the car begins to rotate (roll) about the outer wheels i.e., rollover begins.

The two cases are :-

1. Turning left: friction on the car acts leftward, producing a toppling torque τ_{toppling} pointing toward the back of the car (about the front-back axis).

2. Turning right: friction on the car acts rightward, producing a toppling torque τ_{toppling} pointing toward the front of the car (about the front-back axis).

We will exploit the turning motion (yaw angular velocity) to prevent toppling. If there is a Flywheel with Angular Momentum L_m Perpendicular to the turning Torque, The turning torque produces a yaw angular velocity, and that yaw motion Ω_{yaw} tries to change the direction of the flywheel's angular momentum L_m and if the the Flywheel is fixed at a place, there will be reaction forces to cause Gyroscopic Torque on the car. So we have to assemble the Flywheel in such a way that the Gyroscopic Torque acts opposite to toppling Torque.

With all the above considerations, the direction of L_m should be pointing towards the Lateral Right side Direction of the Automobile. This ensures that $\Omega_{\text{yaw}} \times L_m$ always points opposite to τ_{toppling}

Hence,

**The Flywheel must be placed with its Axis of Rotation Lateral
(left to right) to the Car and spin in the Direction
Opposite to the Direction of wheels**

b.

$$L_m = \frac{mR^2\omega}{2}$$

Let r be the radius of the curve through which the car is turning. So

$$\Omega_{\text{yaw}} = \frac{v}{r}$$

Equal loading on both wheels implies Normal Reaction Forces on both wheels is equal. So if we write the Torque Equation about the C.M. of the Car, the contribution of Normal Forces is Zero. This leaves us with the Torque created by the Force of Friction and the Torque of Flywheel to deal with.

Here

$$\text{Net Frictional Force} = \text{The Centripetal Force} = \frac{Mv^2}{r}$$

So Torque Equilibrium Relation comes out to be

$$\frac{Mv^2}{r}L = \frac{v}{r} \frac{mR^2\omega}{2}$$

Hence,

$$\omega = \frac{2vML}{mR^2}$$

Problem 3

Kleppner & Kolenkow, Problem 7.9

7.9 This problem involves investigating the effect of the angular momentum of a bicycle's wheels on the stability of the bicycle and rider. Assume that the center of mass of the bike and rider is height $2l$ above the ground. Each wheel has mass m , radius l , and moment of inertia ml^2 . The bicycle moves with velocity V in a circular path of radius R . Show that it leans through an angle given by

$$\tan \phi = \frac{V^2}{Rg} \left(1 + \frac{m}{M} \right),$$

where M is the total mass.

The last term in parentheses would be absent if angular momentum were neglected. Do you think that it is important? How important is it for a bike without a rider?

Since the question have not mentioned any Banking of the road, we will consider No Banking. Let the Cycle move around in Anti-Clockwise direction as seen from above and plane of rotation be the xy Plane.

Torque due to Gravity is

$$|\tau_G| = |Mg2l \sin(\phi)|$$

Friction provides the necessary Centripetal Force for the Circular Motion in this case. So Torque due to Friction is

$$|\tau_f| = \left| \frac{MV^2}{R} 2l \cos(\phi) \right|$$

In the absence of Angular Momentum, the Torque Equilibrium gives

$$Mg2l \sin(\phi) - \frac{MV^2}{R} 2l \cos(\phi) = 0$$

$$\Rightarrow \tan(\phi) = \frac{V^2}{Rg}$$

By our Consideration of the Cycle's Motion, we get,

$$\vec{\Omega} = \frac{V}{R} \hat{z}$$

$$\vec{L} = 2ml^2 \frac{V}{l} (-\cos(\phi) \hat{\perp}_{xy} - \sin(\phi) \hat{\parallel}_{xy}) = 2ml(-\cos(\phi) \hat{\perp}_{xy} - \sin(\phi) \hat{\parallel}_{xy})$$

Therefore, Gyroscopic Torque is

$$|\tau_P| = |\vec{\Omega} \times \vec{L}| = \frac{2mlV}{R} \cos(\phi)$$

Including the Angular Momentum, the Torque Equilibrium comes out to be

$$Mg2l \sin(\phi) - \frac{MV^2}{R} 2l \cos(\phi) - \frac{2mlV}{R} \cos(\phi) = 0$$

$$\Rightarrow \boxed{\tan(\phi) = \frac{V^2}{Rg} \left(1 + \frac{m}{M}\right)}$$

Importance of the angular-momentum term

From the result

$$\tan \phi = \frac{V^2}{Rg} \left(1 + \frac{m}{M}\right),$$

the contribution due to angular momentum of the wheels is contained in the factor $\frac{m}{M}$.

For a real bicycle with a rider, the mass of each wheel m is much smaller than the total mass M of the bicycle and rider. Hence $\frac{m}{M} \ll 1$, and the angular-momentum term provides only a small correction to the lean angle. The dominant effect determining the lean is the geometric factor $\frac{V^2}{Rg}$.

For a bicycle without a rider, however, the total mass M is comparable to the wheel mass m , so $\frac{m}{M} \sim 1$. In this case the angular-momentum term is no longer negligible and plays an essential role in stabilizing the bicycle.

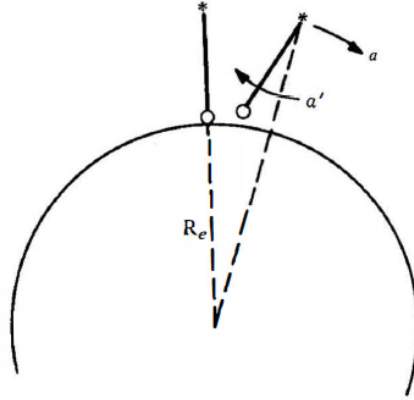
Thus, while angular momentum effects are relatively unimportant for a bicycle with a rider, they become crucial for the stability of a riderless bicycle.

Problem 6

Kleppner & Kolenkow, Problem 8.3

8.3 A pendulum is at rest with its bob pointing toward the center of the earth. The support of the pendulum starts to move horizontally with uniform acceleration a , and the pendulum starts to swing. Find the angular acceleration α' of the pendulum. Find the period of the pendulum for which the bob continues to point toward the center of the earth. Neglect rotation of the earth. This is the principle of a device known as a Schuler pendulum which is used to suspend the gyroscope stage in inertial guidance systems.)

Ans. clue. $T \approx 1\frac{1}{2}$ h



A pendulum is initially at rest with its bob pointing toward the center of the Earth. The support begins to move horizontally with constant acceleration (a). We neglect the rotation of the Earth.

Angular acceleration of the pendulum

It is convenient to analyze the motion in the non-inertial frame attached to the accelerating support. In this frame, the pivot is at rest and the bob experiences, in addition to gravity, a fictitious force due to the acceleration of the frame.

The forces acting on the bob are:

- Gravity ($m\vec{g}$), directed toward the center of the Earth,
- A fictitious force ($-m\vec{a}$), horizontal and opposite the acceleration of the support,
- The tension in the string.

The gravitational and fictitious forces combine to define an *effective gravity*

$$|\vec{g}_{\text{eff}}| = \sqrt{|\vec{g}|^2 + |\vec{a}|^2}$$

Let (δ) be the small angle between the Earth-centered vertical and the direction of $(\vec{g}_{\text{eff}})$. Since $(\vec{g})$ is vertical and $(\vec{a})$ is horizontal,

$$\tan \delta = \frac{a}{g} \quad \delta \simeq \frac{a}{g} \quad (\delta \ll 1)$$

If the pendulum has length (ℓ), the angular acceleration produced by the component of gravity perpendicular to the string is

$$\alpha' = \frac{g}{\ell}, \delta = \frac{g}{\ell} \left(\frac{a}{g} \right) = \frac{a}{\ell}.$$

Thus, the angular acceleration of the pendulum in the accelerating frame is

$$\boxed{\alpha' = \frac{a}{\ell}}$$

Condition for the bob to continue pointing toward Earth's center

As the support moves horizontally over the curved surface of the Earth (*radius*(R_e)), the direction of the local vertical rotates. If the support has horizontal acceleration (a), the angular acceleration of the Earth-centered vertical is purely geometric and given by

$$\alpha_{\text{vertical}} = \frac{a}{R_e}$$

For the bob to continue pointing toward the center of the Earth, the pendulum must acquire exactly the same angular acceleration:

$$\alpha' = \alpha_{\text{vertical}}$$

Using the result from part (a)

$$\frac{a}{\ell} = \frac{a}{R_e}$$

Canceling (a),

$$\boxed{\ell = R_e}$$

Period of the Schuler pendulum

For a simple pendulum of length (ℓ), the small-oscillation period is

$$T = 2\pi \sqrt{\frac{\ell}{g}}.$$

With ($\ell = R_e$),

$$T = 2\pi \sqrt{\frac{R_e}{g}} \approx 84.4 \text{ minutes} \approx 1.4 \text{ hours}$$

Final Results

$$\boxed{\alpha' = \frac{a}{\ell} \quad \ell = R_e \quad T = 2\pi \sqrt{\frac{R_e}{g}} \approx 1.4 \text{ hours}}$$

This special pendulum, whose length equals the Earth's radius, is known as a *Schuler pendulum* and forms the basis of inertial guidance systems.

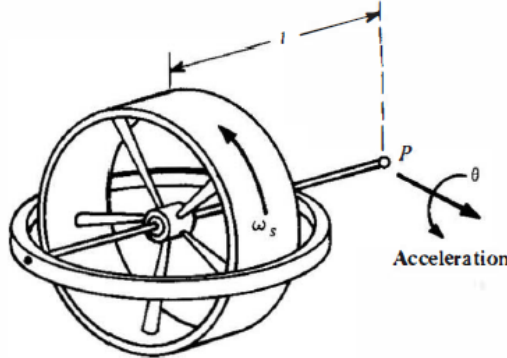
Problem 8

Kleppner & Kolenkow, Problem 8.5

8.5 Many applications for gyroscopes have been found in navigational systems. For instance, gyroscopes can be used to measure acceleration. Consider a gyroscope spinning at high speed ω_s . The gyroscope is attached to a vehicle by a universal pivot P . If the vehicle accelerates in the direction perpendicular to the spin axis at rate a , then the gyroscope will precess about the acceleration axis, as shown in the sketch. The total angle of precession, θ , is measured. Show that if the system starts from rest, the final velocity of the vehicle is given by

$$v = \frac{I_s \omega_s}{Ml} \theta,$$

where $I_s \omega_s$ is the gyroscope's spin angular momentum, M is the total mass of the pivoted portion of the gyroscope, and l is the distance from the pivot to the center of mass. (Such a system is called an integrating gyro, since it automatically integrates the acceleration to give the velocity.)



A gyroscope is spinning rapidly with spin angular speed ω_s and is attached to a vehicle by a universal pivot P . The center of mass of the gyroscope is at a distance l from the pivot. The vehicle accelerates with acceleration a in a direction perpendicular to the spin axis. The total angle of precession θ is measured. The system starts from rest.

It is convenient to analyze the motion in the non-inertial frame of the accelerating vehicle. In this frame, the gyroscope experiences a fictitious force

$$\vec{F}_{\text{pseudo}} = -M\vec{a}$$

acting at its center of mass.

Since the center of mass is a distance l from the pivot, this force produces a

torque about the pivot of magnitude

$$\tau = M a l$$

For a rapidly spinning gyroscope, the change in angular momentum is governed by

$$\boldsymbol{\tau} = \frac{d\vec{L}}{dt} = \boldsymbol{\Omega} \times \vec{L}$$

where $\boldsymbol{\Omega}$ is the precession angular velocity and

$$L = I_s \omega_s$$

is the spin angular momentum of the gyroscope.

Equating the magnitudes,

$$M a l = \Omega I_s \omega_s$$

Now write the acceleration and precession rate as time derivatives:

$$a = \frac{dv}{dt}, \quad \Omega = \frac{d\theta}{dt}$$

Substituting,

$$M l \frac{dv}{dt} = I_s \omega_s \frac{d\theta}{dt}$$

Rearranging,

$$M l dv = I_s \omega_s d\theta$$

Integrating from rest ($v = 0$ when $\theta = 0$),

$$\int_0^v M l dv = \int_0^\theta I_s \omega_s d\theta$$

This yields

$$M l v = I_s \omega_s \theta$$

Therefore, the final velocity of the vehicle is

$$v = \frac{I_s \omega_s}{M l} \theta$$

The precession angle θ is proportional to the time integral of the acceleration, i.e. the velocity. Hence, this device functions as an *integrating gyroscope*.